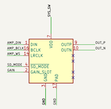
BONE BUDS REV B | AUDIO_LEFT | BARE IC ENGINEERING PROTOTYPE

RF matching, impedance, USB signal integrity, acoustic performance and cell safety require engineering qualification.

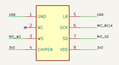
Controller requires protected 1S 4.2V LiPo pack + 10k NTC. USB defaults 100mA; firmware may select 500mA ONLY after enumeration.

Audio: 64 clocks per stereo frame. Separate microphone input and amplifier output I2S buses. BTL speaker terminals must never be grounded.

U1
MAX98357A



U2
CM4-40300-261-I2S-TR



J1
JST SH 10P



R1
2k

R2
100k / 3dB

C1
100n

C2
22u 10V

C3
100n

C4
10u

FB1
BLM18PG121SN1

FB2
BLM18PG121SN1

C5
1n C0G

C6
1n C0G

J2
JST SH 2P

TP1
GND

TP2
SD_MODE

#FLG16
PWR_FLAG

#FLG17
PWR_FLAG

#FLG18
PWR_FLAG

Sheet: /
File: audio_left.kicad_sch

Title: Bone Buds Rev B — audio_left

Size: A2 | Date: 2026-05-23 | Rev: B EVT / qualification

KiCad E.D.A. 10.0.3 | Id: 1/1